

Seismic Hazard Assessment Frameworks in High-Risk Indian Regions: A Methodological Review and Gap Analysis with Reference to Itanagar, India

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Abstract: This paper presents a state-of-the-art review and critical gap analysis of seismic hazard assessment methodologies, focusing on deterministic and probabilistic approaches, ground response modeling, and microzonation techniques. While Itanagar City, Arunachal Pradesh, India, is used as an illustrative example to contextualize regional challenges, this review does not present a new applied seismic hazard assessment for the area. Instead, it highlights current methodological advances, data limitations, and regulatory gaps that influence seismic risk management for high-seismicity urban regions in India. The study also emphasizes the importance of detailed site characterization using geological, geomorphological, geophysical, and geotechnical investigations. Advanced geophysical methods such as multichannel analysis of surface waves and standard penetration tests are used to derive shear-wave velocity profiles, which are vital for understanding site amplification and soil behavior during seismic events. The paper reviews both equivalent-linear and nonlinear ground response analysis techniques, highlighting their strengths and limitations in modeling local soil responses. Additionally, the review underscores the role of seismic microzonation in urban planning and infrastructure development. By integrating seismic hazard assessment with ground response and site-specific data, high-resolution microzonation maps can be developed to guide resilient construction practices and disaster preparedness. The review identifies existing gaps in data quality, attenuation relationships, and regional $G/G_{\max}$ and damping ratio curves, especially for Itanagar. The study concludes that although microzonation initiatives have been carried out in several Indian cities, most remain at a preliminary stage. Continuous refinement through multidisciplinary approaches and region-specific investigations is necessary to transform these maps into effective tools for civil engineers and policymakers. DOI: 10.1061/NHREFO.NHENG-2606. © 2026 American Society of Civil Engineers.

Introduction

Over the historic times, earthquakes have always culminated to be a devastating hazard. Numerous significant earthquakes have struck various regions worldwide, including Armenia (1988, M_s 6.8), Iran (1990, M_w 7.4), the United States (1994, M_w 6.7), Japan (1995, M_w 6.9), Mexico (1985, M_w 8.0), Taiwan (1999, M_w 7.7), India (2001, M_w 7.7), Alaska (2002, M_w 7.9), Sumatra (2004, M_w 5.6), Pakistan (2005, M_w 7.6), China (2008, M_w 7.9), Haiti (2010, M_w 7.0), Japan (2011, M_w 9.0–9.1), Chile (2015, M_w 8.3–8.4), Turkey (2023, M_w 7.8), the Tonga Islands (2025, M_w 7.0), Tibet (2025, M_w 7.1), and Myanmar (2025, M_w 7.7). These devastating occurrences have driven extensive research into seismic hazards, focusing on subsurface engineering attributes, regional tectonic

features, and geological formations as fundamental elements. Consequently, global efforts, such as the Global Seismic Hazard Assessment Program, have facilitated seismic risk evaluations across multiple regions (Giardini et al. 1999). The Indian subcontinent ranks among the most seismically active regions in the world due to its complex tectonic framework. The primary seismogenic zones are linked to the convergent plate boundary where the Indian Plate collides with the Eurasian Plate. This tectonic activity is prominently manifested in the Kirthar-Sulaiman ranges, the Himalayan Mountain belt, and the Arakan-Yoma fold and thrust zone, which delineate the northern and northeastern boundaries of the subcontinent (Parvez et al. 2003). These regions are characterized by frequent earthquakes, varying from moderate to large magnitudes, and are critical for understanding seismic hazard in South Asia. India has a history of experiencing major earthquakes, including Assam (1897, M_w 8.7), Kangra (1905, M_w 8.6), Bihar-Nepal (1934, M_w 8.4), and Assam-Tibet (1950, M_w 8.7). More recently, the nation has endured significant seismic events in Uttarkashi (1991, M_w 6.5), Latur (1993, M_w 6.4), Jabalpur (1997, M_w 6.0), Chamoli (1999, M_w 6.8), Bhuj (2001, M_w 7.7), and Kashmir (2005, M_w 7.4) and the 2011 Sikkim earthquake (M_w 6.9), 2016 Manipur Earthquake (M_w 6.7), 2017 Tripura earthquake (M_w 5.7), 2020 M izoram earthquake (M_w 5.6), 2021 Assam earthquake (M_w 6.4), 2022 Arunachal Pradesh earthquake (M_w 5.7), 2023 Meghalaya (M_w 5.4), and 2025 Manipur earthquake (M_w 5.7). Approximately 60% of India's land area is susceptible to seismic hazards (NDMA 2010). Earthquakes remain unpredictable and pose serious threats, including site deformation, ground displacement, liquefaction, landslides, and tsunamis. The risks are amplified by inadequate urban planning, dense habitations, and substandard construction practices. Seismic strain gaps in

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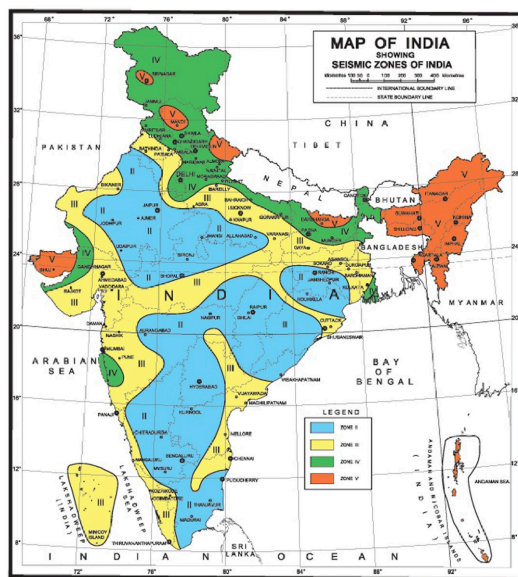
subduction zones further increase the likelihood of large-scale earthquakes. The frequent occurrence of minor tremors in the Himalayan region suggests the potential for a significant earthquake in the central Himalayan seismic strain gap. Despite technological advancements in earthquake-resistant design, further improvements in engineering and construction practices are needed to minimize casualties and structural damage. The rapid growth of urban areas near seismically active regions emphasizes the necessity of seismic microzonation and hazard assessment.

Seismic microzonation entails dividing a region into smaller zones based on localized earthquake effects to aid in new construction planning and the retrofiting of existing buildings. This process assesses soil behavior under seismic activity and evaluates earthquake hazards on a finer scale by mapping and computing impacts. Microzonation maps, which consider local site conditions, are valuable tools for urban development and designing earthquake-resistant infrastructure. Various Indian cities have been analyzed for seismic hazards, including Delhi (Rao and Satyam 2004), Bangalore (Anbazhagan and Sitharam 2008), Imphal (Raghukanth and Nadh Somala 2009), Kolkata (Govindaraju and Bhattacharya 2012), Ahmedabad (Rao et al. 2012), Gujarat (Shukla and Choudhury 2012b), Kanpur (Jishnu et al. 2013; Anbazhagan et al. 2017a), Haryana (Puri and Jain 2019), Mumbai (Desai and Choudhury 2015), Vijayawada (Satyam and Towhata 2016), Surat (Singh et al. 2017), Chennai (Nampally et al. 2018), and Vishakhapatnam (Putti and Satyam 2018). These studies highlight the significance of site-specific ground-motion estimation in ensuring structural safety against earthquakes.

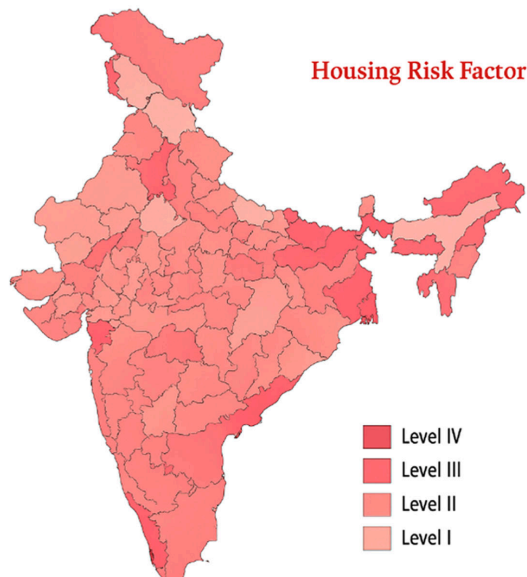
Seismic microzonation involves several analytical approaches, including probabilistic and deterministic seismic hazard evaluations, site characterization, ground response analysis, and liquefaction assessment. Probabilistic seismic hazard analysis (PSHA) estimates earthquake hazards by considering potential seismic sources, their recurrence rates, and the probability of exceeding specific ground-motion levels. Deterministic seismic hazard analysis (DSHA), on the other hand, evaluates hazards based on the maximum plausible earthquake scenario for a particular fault. Both methodologies contribute to a comprehensive understanding of seismic risks and are employed as required. The review by Mihalić

et al. (2011) comprehensively outlines the principles and practices of seismic microzonation, emphasizing its importance in hazard assessment and urban planning. While the paper primarily focuses on traditional parameters such as PGA and S_a , it does not adequately address newer integrative approaches such as PSHA combined with site-specific microzonation techniques. Another limitation is the lack of case studies from regions with heterogeneous subsurface conditions, which could illustrate how microzonation practices adapt to varying geological and geotechnical complexities. Site characterization involves geological, geophysical, and geotechnical studies, utilizing parameters, such as the average seismic shear-wave velocity from the surface to a depth of 30 m (V_{s30}), standard penetration test (SPT) values, and subsurface stratigraphy, to categorize sites according to their seismic response. Site response studies analyze how the subsurface layers react to seismic waves, helping to determine seismic hazard parameters such as the PGA and response spectra, which are crucial for earthquake-resistant infrastructure planning and hazard mitigation.

Walling and Mohanty (2009) reviewed seismic zonation and microzonation studies in India, highlighting advancements from broad macrozonation to detailed site-specific microzonation for urban centers. The study emphasized integrating geophysical, geological, and geotechnical data to assess seismic hazards, with parameters such as PGA and site amplification factors. However, it lacked comprehensive discussions on challenges in data availability and the enforcement of microzonation practices in developing regions. Additionally, the manuscript did not address modern methodologies such as PSHA or advanced numerical modeling, leaving a gap in its applicability to current urban planning needs. These limitations point to the need for further refinement in microzonation approaches for effective hazard mitigation. The Bureau of Indian Standards (BIS) first released the Earthquake Hazard Zonation Map of India in 1962 [IS 1893:1962 (BIS 1962)]. Since then, it has been periodically updated based on significant earthquakes that have occurred in the country. The most recent version was published in 2016 [IS 1893:2016 (Part 1) (BIS 2016)]. According to this map, approximately 56% of India's geographical area is susceptible to moderate to severe earthquake shaking, as shown in Fig. 1(a). A housing risk factor is used to understand the overall status of housing in India



(a)



(b)

Fig. 1. (a) Seismic hazard map; and (b) housing threat factor (after housing risk factor distribution in India). [Reprinted (a) from IS 1893:2016 (Part 1); adapted (b) from Ramancharla and Murty 2014.]

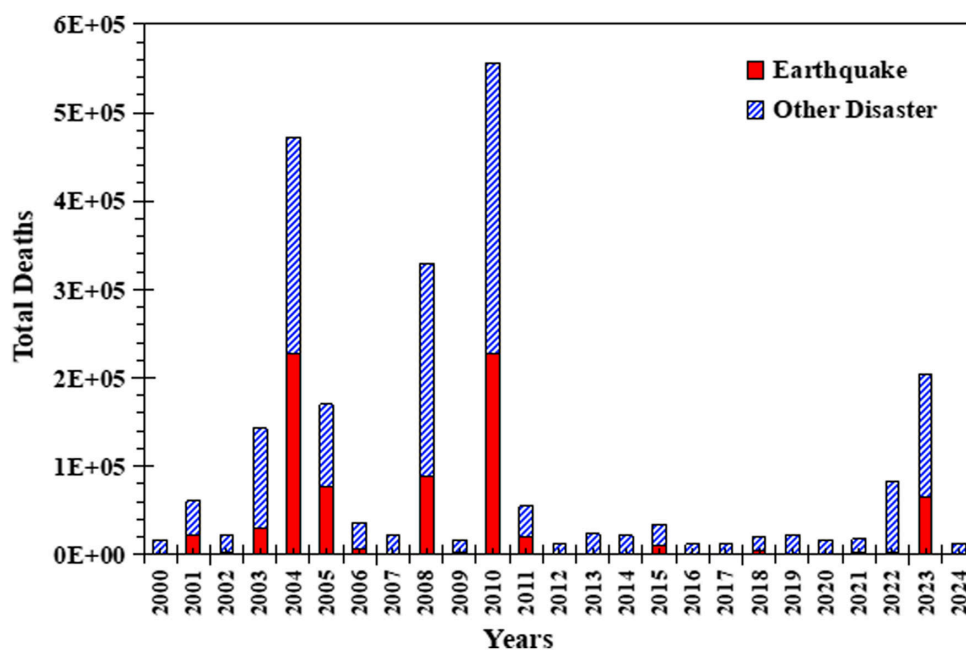


Fig. 2. Worldwide disaster-related deaths (2000–2024) based on EM-DAT (2025) data.

(Ramancharla and Murty 2014). This factor is obtained by multiplying the seismic zone factor by number of houses per km^2 for each district in India in each corresponding seismic zone [Fig. 1(b)]. Fig. 2 illustrates worldwide disaster-related deaths from earthquakes and other disasters between 2000 and 2024, based on data from the Emergency Events Database.

The continuous subduction of Indian plate beneath the Eurasian plate has driven the formation of the Himalayas (Oldham 1883, 1899; Kayal 1987; Kumar et al. 2005; Mukhopadhyay 1992; Mukhopadhyay and Dasgupta 1988; Nandy 2001; Tandon and Srivastava 1975; Mitra et al. 2005) and has consequently led to the development of a seismotectonic zone all along the Himalayan belt. Northeast India ranks among the most seismically active plate tectonic regions globally (Kayal 1998; Tewari 2000). Numerous high-magnitude earthquakes have struck this region, where frequent foreshocks and aftershocks indicate persistent seismic activity. Due to this high seismicity, government regulations limit high-rise construction, impacting regional development. Studies have been conducted for seismic hazard assessment in northeast Indian cities such as Guwahati (Raghukanth 2010; Nath et al. 2008, 2009), Sikkim (Nath et al. 2005, 2008), Manipur (Pallav et al. 2010), Tripura, and Mizoram (Sitharam and Sil 2014).

Despite extensive research in various Indian cities, a comprehensive seismic hazard assessment for Itanagar is yet to be conducted, even though the region is highly prone to seismic shaking and ground amplification. The increasing population, along with rapid and unregulated urbanization, further exacerbates the risk. Itanagar, the capital of Arunachal Pradesh, covers approximately 51.69 km^2 and is located in the outer Himalayas. The area features rugged terrain with elevations ranging from 80 m to 1,540 m and comprises Quaternary fluvial deposits forming river terraces and valley fills. Evidence of past seismic activity is apparent in the Siwalik and Quaternary sediments, reinforcing the need for microzonation. Given that Itanagar falls under the highest seismic risk category, Zone V [IS 1893:2016 (Part 1)], with a zone factor of 0.36 g, the city is particularly vulnerable to earthquake hazards. Conducting a detailed seismic microzonation study is essential for assessing the geological framework, evaluating seismic risks, and implementing earthquake mitigation strategies. This review emphasizes the

necessity of seismic hazard assessment in Itanagar, advocating for a systematic approach to site-specific earthquake hazard assessments. A thorough study would aid policymakers, urban planners, and engineers in constructing earthquake-resilient infrastructure, thereby enhancing regional preparedness for future seismic events. To further contextualize these changes, the evolution of seismic codes and zonation maps over time offers critical insights into the adaptation of seismic standards in response to major earthquakes.

Seismic Zonation Maps of India: Evolution and Revisions

Following the devastating Bihar-Nepal earthquake of 1934, the Geological Survey of India (GSI) created the country's first national seismic hazard map in 1935. In 1947, the Indian Standards Institution was established to set industrial and commercial standards; this body was renamed the BIS in 1986 under the Bureau of Indian Standards Act. BIS is responsible for developing, certifying, and regulating product quality across various sectors, including industry, scientific organizations, regulatory authorities, educational institutions, and government ministries. Importantly, BIS is the national authority designated to prepare and publish seismic hazard maps and design codes in India, which are recognized by other national bodies. In 1962, BIS released the first official seismic zonation map of India (IS 1893:1962), using earthquake epicenter data and isoseismal maps originally produced by GSI in 1935. This map considered earthquakes of magnitude 5 and above, focusing on areas that experienced shaking intensities between V and IX on the Modified Mercalli Intensity (MMI) scale. Based on this, India was divided into seven seismic zones, from Zone 0, indicating minimal or no expected damage, to Zone VI, representing areas with potentially severe damage, as illustrated in Fig. 3(a).

In the initial seismic zonation, the Deccan Plateau was regarded as relatively stable and assigned Zone 0, indicating minimal seismic hazard. In contrast, a significant portion of northeastern India was classified under Zone VI, representing a high-risk area. A revision of this zonation was undertaken in 1966 [IS 1893:1966 (BIS 1966)], incorporating additional parameters such as geological

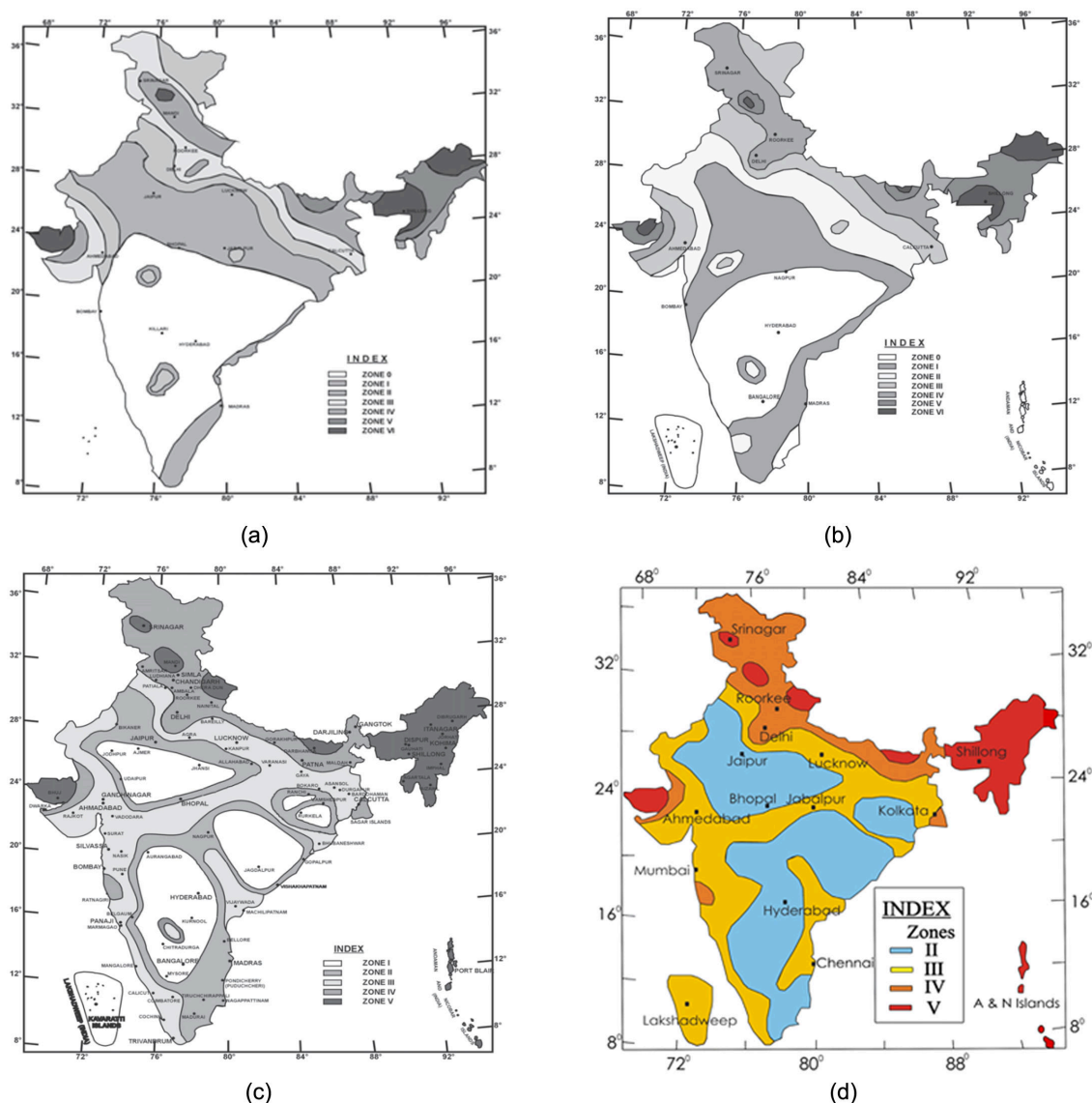


Fig. 3. Seismic zone maps of India: (a) prepared in 1962; (b) prepared in 1966; (c) prepared in 1970; and (d) prepared in 2002. Zones I and II are merged to form four zones. [Reprinted (a) from IS 1893:1962; (b) from IS 1893:1966; (c) from IS 1893:1970; and (d) from IS 1893:2002 (Part 1).]

characteristics and tectonic structures as shown in Fig. 3(b). Tectonic mapping, especially of fault systems (Tandon 1992), played a key role in updating the zonal boundaries. Both the 1962 and 1966 maps divided India into seven zones, each assigned a maximum MMI based on past or expected seismic activity.

A significant revision to the seismic zonation map was carried out in 1970 following the 1967 Koyna earthquake, which had a surface wave magnitude of M_s 6.5 and struck the Deccan Plateau, an area previously classified as Zone 0. This event highlighted the need to incorporate both geological and geophysical evidence in the reassessment of seismic zoning. One of the key changes was the elimination of Zone 0, as it was scientifically unjustifiable to consider any region entirely free from seismic activity. Additionally, Zones V and VI were combined into a single category [IS 1893:1970 (BIS 1970)]. As a result, the updated classification system reduced the number of seismic zones from seven to five [Fig. 3(c)].

The 1993 Latur earthquake, with a body wave magnitude (M_b) of 6.3, resulted in damage intensity reaching IX on the MSK scale. Notably, prior to this event, Latur was categorized under Seismic Zone I, an area where such a strong earthquake was not anticipated.

This unexpected seismic activity was followed by other damaging earthquakes the 1997 Jabalpur (M_b 6.0) and the 2001 Bhuj (M_w 7.6) which further cast doubt on the reliability of the existing seismic zonation for Peninsular India. These events prompted a reassessment of the seismic hazard classification, leading to the publication of a revised seismic zonation map in 2002 [IS 1893:2002 (Part 1) (BIS 2002)], where the number of zones was reduced to four, namely, Zones II, III, IV, and V [Fig. 3(d)].

The seismic zonation map presented in IS 1893:2016 (Part 1), as shown in Fig. 1(a), introduced modification over the earlier IS 1893:2002 (Part 1) version, while continuing with the four-zone classification system with Zones II, III, IV, and V. One of the major improvements in BIS (2016) was the inclusion of updated seismic data and enhanced understanding of tectonic features across India, which resulted in refined and more accurate delineation of seismic zones. Although the zone factors remained consistent with the previous version (ranging from 0.10 in Zone II to 0.36 in Zone V), BIS (2016) adopted a more performance-based design approach, emphasizing the use of dynamic analysis in structural design. Greater attention was given to local site conditions, soil

profiles, and structural irregularities to improve the safety and effectiveness of earthquake-resistant designs.

The changes made to India's seismic zonation maps in response to major earthquakes show that regional-scale hazard assessments often miss localized variations in seismic risk. These maps represent the current understanding of seismic activity and the methods used to assess it. However, they are not definitive and must be regularly updated to include new data, advanced analysis techniques, and deeper insights into local seismic behavior. While macrolevel zonation offers a broad overview, more detailed seismic microzonation studies provide a finer, site-specific perspective. Currently, IS 1893:2016 (Part 1) is being revised to include updated city-level seismic microzonation. This highlights the importance of conducting detailed city-level seismic hazard assessments, which are essential for developing updated zonation maps directly applicable to engineering designs and construction practices.

Seismic Microzonation: Methodological Framework and Key Steps

Seismic microzonation serves as a foundational step in earthquake risk mitigation and relies on a multidisciplinary approach, drawing significant inputs from geology, seismology, geophysics, geotechnical, and structural engineering. A thorough understanding of the tectonic setting and geological formations within the study area is crucial for identifying seismic sources and developing realistic earthquake hazard models. The process demands detailed field investigations to assess seismic hazards at a granular level, effectively capturing spatial variations in ground response. These microzonation studies play a vital role in evaluating potential risk scenarios and are especially beneficial for urban planning, as they aid in forecasting the impact of future earthquakes. Moreover, microzonation maps support informed decisions regarding the placement of critical infrastructure such as hospitals, fire stations, and emergency response centers. They are also instrumental in safeguarding heritage sites and other important structures against damage from future seismic events.

Seismic microzonation is classified into different levels based on the scale of mapping and the depth of investigation involved. According to the Technical Committee on Earthquake Geotechnical Engineering of the International Society of Soil Mechanics and Geotechnical Engineering (TC4-ISSMGE 1999), a Level I microzonation map is considered the most basic and is prepared at a scale ranging from 1:1,000,000 to 1:50,000. It relies primarily on historical earthquake records and available geological and geomorphological data. A Level II map involves a more detailed approach, typically produced at a scale between 1:100,000 and 1:10,000, where ground motion is estimated using microtremor measurements and simplified geotechnical investigations. The most refined form, Level III, includes comprehensive geotechnical investigations and detailed ground response analysis (GRA), with mapping conducted at a finer scale ranging from 1:25,000 to 1:5,000. Each ascending level represents a more rigorous and localized assessment of seismic hazard, offering greater accuracy and relevance for engineering applications. Seismic zonation generally consists of the following three major phases:

- *Earthquake source characterization:* This phase involves identifying and defining the seismic sources using a seismotectonic map. It forms the foundation for both deterministic and probabilistic seismic hazard analyses to estimate ground-motion parameters at the bedrock level.
- *Site characterization:* Detailed assessment of local site conditions is carried out using geological, geomorphological, geophysical, and geotechnical investigations. This phase also

considers influencing factors such as topography, basin geometry, and soil nonlinearity, which are critical for evaluating site-specific seismic responses.

- *Hazard assessment and data interpretation:* The information gathered in the previous phases is analyzed to estimate local seismic hazards such as site-specific seismic responses, ground-motion amplification, and liquefaction potential. This analysis helps in urban development planning and risk mitigation strategies.

Although seismic microzonation effectively identifies and categorizes site-specific seismic hazards, its accuracy and relevance are deeply dependent on the characteristics of the seismic input motion used for analysis. This necessitates a robust seismic hazard assessment, which is typically achieved through either deterministic or probabilistic approaches. The following section delves into the methodologies of PSHA and DSHA, which form the backbone of defining seismic loading for site-specific studies.

Seismic Hazard Assessment: Deterministic (DSHA) and Probabilistic (PSHA) Approaches

To arrive at a seismic hazards assessment of a region, a comprehensive analysis of the tectonics in the seismic study region needs to be conducted, leading to the development of an earthquake catalogue derived from both historical accounts and instrumental measurements. PSHA estimates the likelihood that a site will experience different levels of ground shaking within a given time frame, while explicitly accounting for uncertainties in both earthquake occurrence and ground-motion prediction. In contrast, DSHA evaluates the impact of a specific scenario, usually the maximum credible earthquake on ground shaking at the site. The selection between PSHA and DSHA depends on the purpose of the study and the level of detail required, where PSHA is typically favored for regional-scale hazard assessments and long-term risk planning, whereas DSHA is more commonly applied to site-specific evaluations and the design of critical infrastructure. Using the available tectonic data, a detailed tectonic map of the region is created, highlighting active tectonic features by superimposing earthquake epicenters from the catalogue onto the map. The maximum magnitude potential of identified seismogenic sources is calculated, incorporating factors such as the total fault length, rupture length, and highest observed magnitudes. Additionally, seismicity parameters are derived to estimate return periods for anticipated earthquake magnitudes, contributing valuable insights to seismic risk assessment. Following a comprehensive literature review, an appropriate attenuation relationship, such as a ground-motion prediction equation (GMPE), is identified to compute PGA and S_a . Hazard maps are lastly generated to illustrate PGA and S_a values across the region for different return periods and exceedance probabilities. Response spectra are developed for key urban centers of cities, facilitating structural engineers and designers in the development of earthquake-resistant structures. The region is then systematically classified into seismogenic zones based on the calculated PGA values. A meticulously compiled earthquake catalogue will underpin these efforts, serving as a statistically reliable resource for robust seismic hazard evaluation. Schematic workflow of the PSHA framework shown in Fig. 4, showing the sequential steps beginning with preliminary studies, earthquake catalog development, homogenization and declustering, completeness analysis, identification of seismogenic sources, and estimation of maximum magnitude potential and hazard parameters, leading to the execution of PSHA and preparation of seismic hazard maps at the regional/city level.

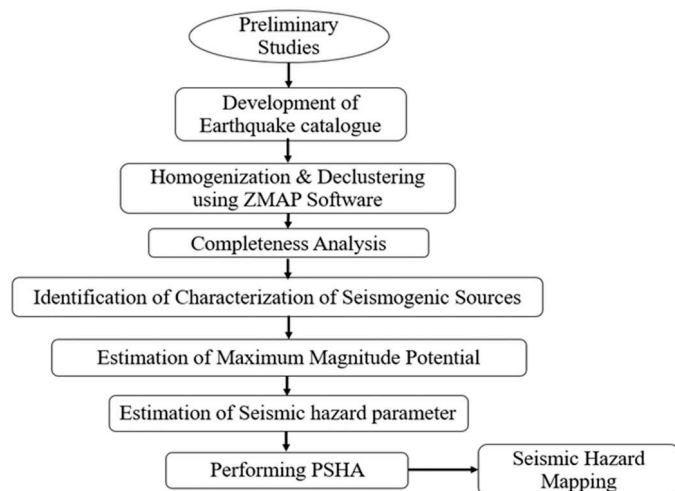


Fig. 4. Flowchart representing the methodological framework adopted for probabilistic seismic hazard assessment (PSHA).

Homogenization, Declustering, and Completeness Analysis of Earthquake Data

Earthquake data are classified into two categories: preinstrumental and instrumental records. Preinstrumental records, derived from historical sources, play a vital role in hazard analysis and are documented in resources such as the National Disaster Management Authority (NDMA) catalogue (NDMA 2010). In contrast, instrumental data are gathered by modern seismological networks sourced from both national and international organizations, including the NDMA, the India Meteorological Department, the International Seismological Center, and the United States Geological Survey (USGS). To ensure the reliability and accuracy of the earthquake catalogue, it undergoes processes such as homogenization, declustering, and completeness analysis. Homogenization involves standardizing earthquake magnitudes as various agencies report data using different magnitude scales, including the moment magnitude (M_w), body wave magnitude (M_b), surface wave magnitude (M_s), local magnitude (M_L), and duration magnitude (M_d). Among these, M_w is preferred due to its reliability at higher magnitudes, as it avoids the saturation issues common with other scales. To create a uniform catalogue, magnitudes from different scales are converted into M_w using the correlation equations [Eqs. (1)–(3)] developed by Scordilis (2006). This standardization is essential to address inconsistencies and ensure compatibility across datasets

$$M_w = 0.85(\pm 0.04)M_b + 1.03(\pm 0.23) \quad 3.5 \leq M_b \leq 6.2 \quad (1)$$

$$M_w = 0.67(\pm 0.005)M_s + 2.07(\pm 0.03) \quad 3.0 \leq M_s \leq 6.1 \quad (2)$$

$$M_w = 0.99(\pm 0.02)M_s + 0.08(\pm 0.13) \quad 6.2 \leq M_s \leq 8.2 \quad (3)$$

To convert local magnitude (M_L) to moment magnitude (M_w), regional correlations are typically considered more reliable. For India and its surrounding regions, the correlation proposed by Kolathayar et al. (2012) is adopted [Eq. (4)]. This region-specific correlation ensures that the conversion best approximates the seismic characteristics of the area, improving the precision of the resulting homogenous earthquake catalogue

$$M_w = 0.815(\pm 0.04)M_L + 0.767(\pm 0.174) \quad 3.3 \leq M_L \leq 7.0 \quad (4)$$

Declustering is another crucial step in ensuring a Poisson distribution of earthquake occurrences by removing dependent events such as foreshocks and aftershocks, which can distort seismic hazard assessments. Various methods for declustering have been proposed, including those by Gardner and Knopoff (1974) and Reasenberg (1985). Additionally, completeness checking is vital to ensure a statistically reliable dataset. Since instrumental earthquake recordings began globally around 1900 and in India in 1960, completeness analysis helps determine whether the dataset sufficiently represents seismic activity over time. The completeness of the catalogue is assessed using methods such as the cumulative visual inspection method (Tinti and Mulargia 1985) and Stepp's method (Stepp 1972). These processes enhance the earthquake database, transforming it into a critical resource for precise seismic hazard assessment and effective earthquake risk mitigation.

Characterization of Earthquake Sources

The foremost step in seismic hazard analysis is the identification and characterization of earthquake sources within and around the study region for which the hazard is to be estimated. The regions that have historically experienced seismic activity and are capable of generating future earthquakes are defined as potential seismic sources. Delineating these source zones is a complex task, as it requires careful consideration of multiple seismological and tectonic parameters, including the fault geometry and mechanism, focal depth, maximum observed magnitudes (M_{obs}), maximum magnitude potential (M_{max}), recurrence rates, and regional tectonic setting. Depending on the available information and scale of analysis, these sources may be represented as points, fault lines, or broader seismic source zones.

Fig. 5 has been created based on the data provided by the Geological Survey of India (GSI 2000) and USGS to produce the seismotectonic map of the study area (i.e., Itanagar, Arunachal Pradesh). The main frontal thrust (MFT), main boundary thrust (MBT), and main central thrust (MCT) are the major south to north running faults in the northeastern region the northern section of the area. These faults are typified with north-dipping listric thrust geometry. The MFT, also referred to as the Himalayan frontal thrust, is the last thrust of the Himalayas, which serves as a boundary between the Brahmaputra basin and the Outer Sub-Himalayas (Fig. 5). Farther north, the MBT divides the Outer Sub-Himalayas and the Lesser Himalayas, which are divided on the north side by the MCT (Kayal 2008; Gupta and Biswas 2022). The major seismotectonic units in the area are the Mishmi Thrust, Lohit Thrust, and right-lateral Po-Chu Fault, which have a NW-SE orientation (Fig. 5). The Naga Thrust and Disang Thrust, NE-SW trending, are major structures in the central section of the region. These are the thrusts that outline the southeastern edge of the Brahmaputra Valley plains in this area. The existence of major thrusts such as the MCT, MFT, Lohit Thrust, and Mishmi Thrust are major contributors to the high seismic activity recorded in the districts of Arunachal Pradesh and the region in general (Das et al. 2006).

Evaluation of Seismicity Parameters

The seismic activity of any fault region can be quantified using two key parameters: a and b , which describe the relationship between moment magnitude (M_w) and the cumulative frequency of occurrence, as introduced by Gutenberg and Richter (1954) and shown using Eq. (5)

$$\log \lambda = a - bM \quad (5)$$

In this framework, multiple alternative models are defined for key components of the analysis, such as source characterization, recurrence parameters, and GMPEs (Maiti et al. 2017). Each branch of the tree represents a discrete model or hypothesis, and a weight is assigned to reflect its relative credibility based on prior research or expert judgment. The weights propagate through the branches such that the sum of the final probabilities equals unity, ensuring a consistent probabilistic treatment (Anbazhagan et al. 2019). Site effects were incorporated through the use of shear-wave velocity profiles obtained from multichannel analysis of surface waves (MASW) and SPT data, which were subsequently used to determine V_{s30} values and classify sites according to NEHRP (2020) guidelines. These classifications were integrated into the selected GMPEs, which explicitly account for site terms (e.g., rock, stiff soil, or soft soil conditions). This ensured that local soil conditions were reflected within the hazard estimates, thereby improving the realism of the ground-motion predictions.

Hazard Curve Calculation

The final output of PSHA is typically expressed in the form of hazard curves, which describe the annual frequency of exceedance for different levels of ground-motion intensity measures (e.g., PGA, spectral acceleration). These curves are derived from the fundamental hazard integral (Cornell 1968; Kramer 1996)

$$v_Y(y) = \sum_{i=1}^{N_s} \int_{M_{\min}}^{M_{\max}} \int_{R_{\min}}^{R_{\max}} P(Y > y | m, r, i) f_{M_i}(m) f_{R_i}(r) dm dr \quad (7)$$

where $v_Y(y)$ = annual rate of exceedance of ground-motion level y ; $P(Y > y | m, r, i)$ = probability of exceeding y for an earthquake of magnitude m at distance r from source i , obtained using the selected GMPEs; and $f_{M_i}(m)$ and $f_{R_i}(r)$ = probability density functions of magnitude and distance for source i .

The hazard calculations were performed within the framework of PSHA, originally proposed by Cornell (1968) and later advanced through computational tools such as EQRISK by McGuire (1976), FRISK (McGuire 1978), FRISK 88M (McGuire 2001), OpenSHA (Field et al. 2003), OPENQUAKE (Pagani et al. 2014), and R-CRISIS (Ordaz and Salgado-Gálvez 2017). By combining contributions from all seismic sources, weighted GMPEs, and site classifications through the logic-tree framework, a family of hazard curves is obtained. These curves are then used to estimate ground-motion values corresponding to specific probability levels, such as 10% probability of exceedance in 50 years (return period of 475 years) or 2% probability of exceedance in 50 years (return period of 2,475 years), which are commonly adopted in engineering design codes and seismic microzonation studies (Bahuguna and Sil 2020; Mishra et al. 2024). For the Itanagar case study, these hazard curves provide a quantitative basis for assessing seismic risk in the region. They not only reflect the complex tectonic environment of the northeastern Himalayas but also incorporate the city's site-specific soil conditions, thereby offering valuable input for urban planning, seismic microzonation, and the development of earthquake-resistant infrastructure.

Comparative Evaluation of Seismic Hazard Assessment: Itanagar City versus Other Indian Cities

Seismic hazard assessment methodologies, including DSHA and PSHA, have been widely and effectively applied in major Indian urban cities such as Delhi, Gujarat, Chennai, Bangalore, Kolkata, Manipur, and Guwahati. As these cities have experienced several minor to major seismic intensities, the detailed geotechnical and seismotectonic data are suggested for GRA and microzonation studies to provide more comprehensive understanding of earthquake

risks and also to support resilient urban planning. In contrast, Itanagar city, despite being located in the severe seismic hazard zone (Zone V), has comparatively limited detailed SHA studies. Beyond the regional seismotectonic mapping and some preliminary ground response characterizations, Itanagar lacks in comprehensive deterministic or probabilistic seismic hazard assessments and refined microzonation study. This gap is primarily due to the unavailability of high-resolution site-specific seismo-geotechnical data (such as regionally recoded or synthetically generated strong motion, site/region-specific GMPEs, borehole profiles, strain-dependent dynamic soil characteristics, and shear-wave velocity profiles) and incomplete earthquake catalog (reflecting insufficient instrumental seismic records for the area). In addition, there are still gaps in practical application. These include the lack of updated microzonation maps that combine geological, geotechnical, and seismic data, the limited use of nonlinear ground response models for site-specific hazard analysis, and the insufficient integration of seismic hazard findings into local building regulations and construction practices. Bridging these gaps requires focused and sustained research efforts in localized data acquisition, the development of Itanagar-specific seismic hazard models, and enhanced collaboration between academic researchers, regulatory agencies, and local authorities. Addressing these critical challenges is essential for improving earthquake risk mitigation and enabling informed, risk-sensitive urban development in Itanagar. In India, several important studies have applied PSHA, as summarized in Table 1, demonstrating its relevance to local seismic risk assessments. While microzonation offers spatial granularity in hazard zoning, a robust understanding of seismic input motion often derived through PSHA or DSHA is essential for reliable site-specific analysis. This foundational requirement leads us to site characterization techniques, which are explored in the following sections.

Engineering and Planning Implications of Hazard Results

The hazard curves and maps generated in the study hold significant implications for engineering practice and urban resilience in Itanagar. The PSHA-derived ground-motion values provide the foundation for developing site-specific seismic design spectra, enabling structural designs that better reflect local hazard conditions than the generalized IS 1893:2016 zonal spectra. The amplification patterns identified through GRA emphasize areas where strict enforcement of seismic design codes is essential and where retrofitting of vulnerable and lifeline structures should be prioritized. At the planning scale, the hazard maps highlight zones of differential amplification that can inform land-use and zoning decisions, ensuring that critical facilities such as hospitals, schools, and emergency services are located in safer areas with lower hazard amplification. Beyond design and zoning, these findings contribute to emergency preparedness and resilience planning by identifying neighborhoods most exposed to strong shaking, thereby guiding the prepositioning of evacuation routes, temporary shelters, and emergency service hubs. By framing hazard outputs in terms of their application to design spectra, building codes, land-use planning, and resilience strategies, this study demonstrates that seismic hazard assessment can serve not only as a research exercise but also as a decision-support tool for risk reduction and sustainable urban development in Itanagar and other high-risk Indian cities.

Site Characterization: Implications for Seismic Hazard Studies

Site characterization for seismic hazard studies crucially depends on various in situ field tests used for site characterization. For site

Table 1. Overview of significant seismic hazard studies conducted in India

Region	Seismic zone as per IS 1893:2016 (Part 1)	PGA (g) as per IS 1893:2016 (Part 1)	PGA (g) as per seismic hazard analysis	Reference
Chennai	III	0.16	0.106	Boominathan et al. (2008)
Bangalore	II	0.1	0.121	Anbazhagan et al. (2009)
South India	II–IV	0.1–0.24	0.37	Vipin et al. (2009)
NW Himalaya and its adjoining area	IV to V	0.24–0.36	0.50–0.7	Mahajan et al. (2010)
Gujarat	III to V	0.16–0.36	0.05–0.92	Chopra et al. (2012)
Kolkata	III	0.16	0.12–0.6	Shiuly and Narayan (2012)
Surat	III	0.16	0.1	Thaker et al. (2012)
Andaman and Nicobar	IV	0.24	0.27–0.58	Kataria et al. (2013)
Himachal Pradesh and adjoining areas	IV to V	0.24–0.36	0.09–0.26	Patil et al. (2014)
Delhi-NCR	IV	0.24	0.15–0.23	Rao and Rathod (2014)
Tripura and Mizoram	V	0.36	0.03–0.54	Sitharam and Sil (2014)
Patna	IV	0.24	0.03–0.44	Anbazhagan et al. (2015)
Goa	III	0.16	0.33–0.62	Naik and Choudhury (2015)
Agartala	V	0.36	0.18–0.46	Sil and Sitharam (2016)
West Bengal	II to V	0.1–0.36	0.11–0.45	Maiti et al. (2017)
Guwahati	V	0.36	0.46–0.92	Bahuguna and Sil (2020)
Shillong	V	0.36	0.61–0.83	Agrawal et al. (2023)
Itanagar	V	0.36	0.18–0.36	Anshu et al. (2025b)

characterization, the most frequently used tests include the SPT, cone penetration test (CPT), seismic cone penetration test (SCPT), spectral analysis of surface waves (SASW), P–S logging, and the MASW. A key parameter commonly adopted for site classification is the average shear-wave velocity of the top 30 m (V_{s30}). When direct shear-wave velocity measurements are not available, V_{s30} can be inferred from SPT and CPT data through established empirical relationships, and these derived values are then applied for site classification. General geotechnical investigations typically involve drilling boreholes, sampling at specific points, and conducting in situ or laboratory tests. This approach is suitable for exploring smaller spatial extents of soil and rock medium. However, seismic microzonation requires large-scale exploration, which tends to become infeasible and impractical using conventional geotechnical methods. Geophysical techniques address this limitation by offering a more efficient alternative while also overcoming other challenges associated with traditional investigation methods. A variety of geophysical methods are now available, many of which can generate continuous subsurface profiles. Some methods also provide information on the stiffness properties of the ground, which is particularly beneficial for seismic microzonation. Additionally, geophysical techniques assist in identifying subsurface features such as cavities, backfilled mine shafts, faults, and other discontinuities. In seismic microzonation, obtaining detailed subsurface profiles across large regions is essential. Conventional geotechnical exploration over such expansive areas is both logistically challenging and prohibitively expensive. Geophysical methods serve as the only viable alternative, allowing for the efficient assessment of lateral variability in near-surface materials while minimizing costs. These methods enable the assessment of lateral variability in near-surface materials beneath a site. The SASW technique (Stokoe et al. 1989; Foti 2000) and the MASW approach (Park et al. 1999; Xia et al. 1999) were developed to evaluate shear-wave velocity profiles by utilizing the propagation characteristics of surface waves. The shear-wave velocity structure is a crucial parameter in geotechnical earthquake engineering and has become increasingly common over

the times (Xia et al. 2004; Socco and Strobbia 2004; Foti and Butcher 2004). While both SASW and MASW can be used to obtain shear-wave velocity profiles, MASW is generally preferred due to its efficiency. SASW tests, which rely on only two geophones, are more labor-intensive and less practical for larger explorations.

The equipment used for active MASW surveys comprises an energy source, receivers, and a data acquisition system (DAQ) to facilitate efficient data collection and processing. The energy source produces controlled seismic waves, which are detected by geophones arranged in a specific configuration. Two primary types of energy sources are utilized: impulsive sources (e.g., hammers, weight drops) and vibrating sources (e.g., electromechanical vibrators). In this particular study, a 10 kg sledgehammer with an aluminum base plate served as the impact source, chosen for its practicality and effectiveness in generating high-frequency waves. The geophones, which act as receivers, convert mechanical vibrations into electrical signals. For reliable data acquisition, 4.5 Hz geophones were employed. The DAQ seismograph processed the captured signals by filtering and digitizing the waveforms for further analysis. The aluminum striker/base plate improved the transmission of high-frequency components, ensuring efficient energy transfer. Additionally, seismic cables with evenly spaced connection points connected the geophones to the DAQ system, enabling seamless and efficient data transfer. Fig. 6 shows a schematic diagram depicting the typical field setup and data acquisition process for an active MASW test. A systematic, step-by-step data processing approach is used to isolate surface wave signals from the recorded seismic data. To determine the subsurface soil characteristics in terms of the shear-wave velocity (V_s) profile, the field data are processed to produce a dispersion curve, which is subsequently analyzed through an inversion procedure. To improve data clarity and reduce noise interference, techniques such as temporal muting and frequency filtering are employed (Taipodia and Dey 2017a, b, 2018; Taipodia et al. 2017, 2020). These steps ensure the generation of high-resolution dispersion images (Park et al. 1998; Taipodia et al. 2017b, 2018). The refined images enable the extraction of the

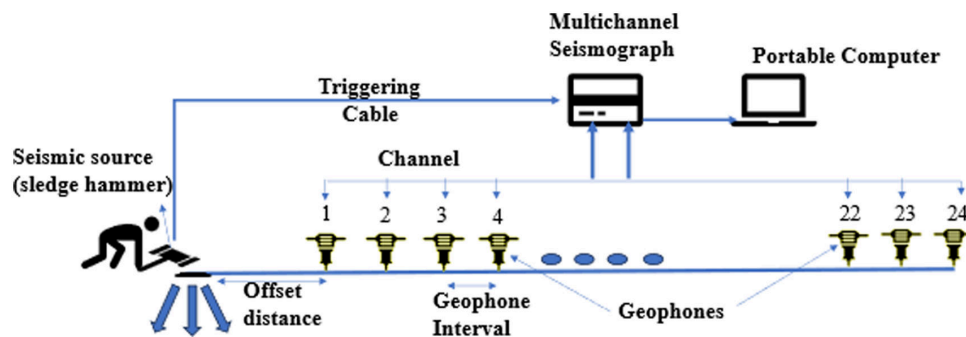


Fig. 6. Schematic diagram of a typical field experimental setup and data acquisition for active MASW test.

fundamental mode dispersion curve having maximum local energy, which is critical for further analysis (Taipodia et al. 2019, 2020). The relation between phase velocities and frequency variations in the fundamental mode dispersion curve is evaluated, with the extracted curve serving as the primary input for the final inversion analysis. This process applies the inversion method developed by Xia et al. (1999), relying on the fundamental mode dispersion curve to generate the inverted shear-wave velocity (SWV) profile at varying depths. The SWV profile is a vital component in geotechnical and engineering practices, as it provides critical details about the stiffness properties of the subsurface soil, supporting the assessment of foundations and structural designs.

Seismic site classification relies on the properties of soil within a depth of 30 m, as recommended by the National Earthquake Hazard Reduction Program (NEHRP 2020) and the International Building Code (IBC 2000, 2006). These classification systems are widely recognized by the international scientific community and serve as the foundation for seismic design manuals (NEHRP BSSC 2000; Dobry et al. 2000; Kanli et al. 2006). According to these guidelines, site classification is determined based on parameters such as shear-wave velocity (V_s), SPT- N values, and undrained shear strength. The role of V_s in site characterization and response studies has been extensively highlighted by researchers (Seed et al. 1976; Idriss 1990, 2008; Boore and Joyner 1997; Joyner and Boore 1981; Joyner and Fumal 1984; Presti et al. 2004). The seismic site classification system utilizes the average shear-wave velocity within the top 30 m of subsurface soil layers (Borcherdt 1994), as defined by Eq. (8)

$$V_{s30} = \frac{30}{\sum_{i=1}^N \left(\frac{h_i}{v_{si}} \right)} \quad (8)$$

where h_i = thickness of individual soil layers up to the cumulative depth; v_{si} = shear-wave velocity of the corresponding individual layers; and N = number of individual soil layers within the depth of 30 m. For seismic site classification, NEHRP (2020) has suggested to consider the SWV of the soil within the depth of 30 m (i.e., V_{s30}) from surface level for characterizing the seismic site response and identifying the seismic parameters required for the design of earthquake resilient structures at regional scale. According to the updated guidelines (NEHRP 2020), depending on V_{s30} , apart from soil sites being classified into six classes as earlier (namely site class A, B, C, D, E, and F), an additional three new site classes (BC, CD, and DE) are introduced to provide better insight of the site shear-wave velocity classes as shown in Table 2.

Mahajan et al. (2007) utilized the MASW test to analyze variations in vertical and horizontal shear-wave velocities in the Dehradun region. According to the NEHRP guidelines (Table 2), Dehradun was classified as Class D and Class C based on average shear-wave velocity. Additionally, spectral acceleration plots for

Table 2. Site classification based on site-specific conditions established by NEHRP (2020)

Site class	Soil profile description	NEHRP (2020) V_{s30} (m/s)
A	Hard rock	>1,500
B	Medium hard rock	>900 to 1,500
BC	Soft rock	>700 to 900
C	Very dense sand or hard clay	>480 to 700
CD	Dense sand or very stiff clay	>330 to 480
D	Medium dense sand or stiff clay	>230 to 330
DE	Loose sand or medium stiff clay	>170 to 230
E	Very loose sand or soft clay	<170
F	Soils requiring site response analysis as per Section 21.1	Ref. Section 20.2.1

Dehradun city were proposed for various time periods and a damping ratio of 5%. In the Bangalore Mahanagar Palika (BMP) region, Anbazhagan and Sitharam (2008) conducted site classification based on the average shear-wave velocity profile. Most of the BMP area was categorized as Class C and Class D according to NEHRP site classification, with the profiles generated from 58 MASW tests. Similarly, Satyam and Rao (2008) performed seismic site characterization for Delhi using the MASW technique at 118 locations. The V_{s30} values for areas such as Jawaharlal Nehru University, Vasanth Kunj, Ladha Sarai, Jarera, and Nangloi Sayied ranged between 400 and 480 m/s, with stiff rock layers identified up to 30 m in depth. Hanumantharao and Ramana (2009) applied the SASW test to quantify shear-wave velocity profiles at 80 sites in the Delhi region. They further developed correlations between V_s and SPT- N values for the same region. To supplement their findings, cyclic triaxial tests were performed using remolded samples of a sand-silt mixture, producing reduction modulus and damping curves. Maheshwari et al. (2010) utilized data from 300 boreholes and conducted 20 MASW tests in Chennai to establish correlations between shear-wave velocity (V_s) with both uncorrected and energy-corrected SPT- N values. Furthermore, a site classification map for Chennai was developed based on the NEHRP guidelines, and the fundamental natural period of different structures was estimated through an empirical correlation between the number of stories and the corresponding time period. It was noted that low-rise buildings in Chennai could be at risk during earthquakes. For Mumbai, Mhaske and Choudhary (2011) developed an average V_s model based on data from 400 boreholes and compared it with existing correlations from other Indian cities. Similarly, Anbazhagan et al. (2013a) completed seismic site classifications for Lucknow using NEHRP guidelines. To establish a correlation between V_s and SPT- N values for the Indo-Gangetic Basin,

47 MASW tests were conducted along with the drilling of 23 boreholes. Based on NEHRP site classification, the majority of Lucknow was categorized as Class D, with some locations falling under Class C. In the city of Agartala, geophysical surveys were conducted at 27 sites using MASW technology (Sil and Sitharam 2014). A quantile–quantile (Q–Q) plot was used to establish correlations between shear-wave velocity (V_s) and SPT-N values, which were subsequently compared with previously reported correlations for the Agartala region. Based on the NEHRP site classification, Agartala was primarily identified as belonging to Class D or Class E, depending on the V_s and SPT-N characteristics. In a related study, Thokchom et al. (2017) performed MASW surveys at 42 locations and P–S logging at 16 sites in the Dholera region of Gujarat to evaluate the shear-wave velocity (V_s). The MASW method was also employed to develop a shear-wave velocity model and perform site characterization for the Kumaon Himalayas (Joshi and Bhardwaj 2018), where the predominant frequency was estimated to range between 1.01 and 5.94 Hz. Similarly, Chakraborty et al. (2018) carried out MASW-based geophysical investigations in Jaipur to estimate one-dimensional (1D) shear-wave velocity profiles, finding that the studied areas could be classified as Class C or Class D under NEHRP guidelines. Parhi et al. (2020) used MASW to assess two ash pond sites in India in terms of shear-wave velocity profiles. For the abandoned ash pond at NTPC Korba, the average V_s values were 149 m/s at the surface and 203 m/s at a depth of 20 m. For the abandoned pond at VTPS Vijayawada, the corresponding values were 161 and 245 m/s. Nagamani et al. (2020) conducted a site characterization study in Gujarat's southern mainland region, estimating subsurface shear velocity profiles up to a depth of 100 m using the ellipticity inversion technique. Based on the geotechnical data, the V_{s30} values for the area were found to range between 520 and 1,350 m/s. In the city of Amaravati, Kumar et al. (2022) presented a correlation between shear-wave velocity and SPT-N values. The study utilized SPT data from 400 boreholes with investigation depths of 25–45 m, alongside data from 30 microtremor stations. The findings indicated that, following NEHRP BSSC (2003) guidelines, the Amaravati region is classified as Class D. The study by Ansari et al. (2024) focused on detailed seismic microzonation of the Jammu region, utilizing DSHA and PSHA to delineate zones of varying seismic risks based on parameters such as shear-wave velocity (V_s) and predominant frequency. Extensive geophysical testing at 242 sites revealed three hazard zones (i.e., low, moderate, and high), which were used to assess the vulnerability of the 345-km-long Udhampur Srinagar Baramulla Rail Link railway project. Aas and Sinha (2023) conducted seismic site characterization at 12 locations in the southern foothills of the tectonically active Himalayan region using MASW data. The V_{s30} values for the studied area were found to range between 285 and 375 m/s. Based on these shear-wave velocity values, the entire foothills region was classified as Site Class C or Site Class D under NEHRP guidelines. A summary of site classification for various Indian cities, determined through MASW investigations and NEHRP standards, is provided in Table 3. The parameters derived from detailed site characterization serve as critical inputs for GRA, enabling the evaluation of seismic wave amplification and soil behavior under earthquake loading, as discussed in the next section.

Ground Response Analysis: Equivalent-Linear and Nonlinear Approaches

In geotechnical earthquake engineering, one of the most essential and commonly required assessments is the evaluation of ground response during seismic events. This analysis is carried out to investigate the amplification or attenuation of seismic

Table 3. Summary of site characterization for Indian cities based on SWV from MASW tests

Reference	Equipment	City	V_s /site classes
Satyam and Rao (2008)	MASW	Delhi	V_s : 185 to 495 m/s
Mahajan (2009)	MASW	Dehradun	Site class: C, D, and E
Kumar et al. (2010)	MASW	Lucknow	Site class: C and D
Maheswari et al. (2010)	MASW	Chennai	Site class: D
Rao et al. (2011)	MASW	Jabalpur	V_s : 250 to 750 m/s
Sil and Sitharam (2014)	MASW	Agartala	Site class: D and E
Chakraborty et al. (2018)	MASW	Jaipur	Site class: C and D
Sairam et al. (2018)	MASW	Ahmedabad	V_s : 260–360 m/s; site class: D
Saha et al. (2021)	MASW	Silchar	Site class: D
Singh et al. (2021)	MASW	Varanasi	V_s : 310 to 690 m/s; site class: C and D
Imam et al. (2023)	MASW	Jamshedpur	V_s : 402 m/s, site class: C

waves, develop response spectra, identify peak ground accelerations and their associated resonant frequencies, evaluate deformation characteristics for assessing liquefaction potential, and estimate earthquake-induced loading on embankment dams. Several methods are available to estimate wave amplification, including linear, equivalent-linear, and nonlinear analyses, which can be conducted in 1D, two-dimensional (2D), or three-dimensional (3D) models. Schnabel et al. (1972) introduced an equivalent-linear approach that approximates the nonlinear hysteretic behavior of cyclically loaded soils, leading to the development of the SHAKE program. This program is commonly applied to perform for 1D equivalent-linear GRA. However, since soil exhibits nonlinear behavior, its shear modulus continuously changes under cyclic loading. The inability of the equivalent-linear model to accurately represent the real-time variations in soil stiffness has been pointed out by multiple researchers, including Liam Finn et al. (1978), Arslan and Siyahi (2006), Hosseini and Pajouh (2012), and Kramer (1996). Studies have demonstrated that nonlinear models provide a more accurate representation of soil behavior during earthquake ground motions (Hosseini and Pajouh 2012). Additionally, for sites characterized by deep soft soil layers or those likely to experience strong earthquakes, the equivalent-linear model is considered less reliable (Hashash et al. 2010). Consequently, most researchers prefer the nonlinear approach for GRA. Widely used nonlinear models have been developed by several researchers in the past (Ramberg and Osgood 1943; Matasović and Vucetic 1993; Hashash and Park 2001). The evaluation of seismic wave amplification is most commonly assessed through a one-dimensional model, which assumes that the seismic waves that are created in the bedrock propagate vertically through the soil layers that are formed on top of the bedrock. But in cases where two- or even three-dimensional modeling is required, the constitutive formulations advanced by Mroz (1967), Momen and Ghaboussi (1982), Dafalias (1986), Kabilamany and Ishihara (1990), Gutierrez et al. (1993), and Cubrinovski and Ishihara (1998) are typically used.

Due to its simplicity, equivalent-linear analysis has been extensively utilized across India for ground response studies. Notable examples include the GRA of Bhuj (Sitharam and Govindaraju 2004), Chennai (Boominathan et al. 2008), Bangalore

Table 4. Overview of previous studies on site response analysis of Indian cities

Region	Seismic zone as per IS 1893:2016 (Part 1)	PGA (g) as per IS 1893:2016 (Part 1)	Maximum amplification factor	Calculated PGA (g)	Reference
Bangalore	II	0.1	1.0–4.90	0.01–0.66	Anbazhagan and Sitharam (2009)
Kutch (Gujarat)	V	0.36	2.45	0.22	Thaker et al. (2010)
Kolkata	III	0.16	1.88–2.07	0.22–0.27	Govindaraju and Bhattacharya (2012)
Surat (Gujarat)	III	0.16	2.1	0.21	Thaker et al. (2012)
Panjim City	III	0.16	1.56–2.36	0.17–0.25	Naik and Choudhury (2013)
Guwahati City	V	0.36	1.2–1.7	0.2–0.79	Kumar and Krishna (2013)
Delhi	IV	0.24	1.8	0.23–0.27	Rao and Rathod (2014)
Mumbai	III	0.16	1.0–4.0	0.07–0.213	Desai and Choudhury (2015)
Vijayawada	III	0.16	0.71	0.27	Satyam and Towhata (2016)
West Bengal	II–V	0.1–0.36	—	0.22–0.77	Maiti et al. (2017)
Itanagar	V	0.36	1.051–4.356	0.218–1.853	Anshu et al. (2024)

(Anbazhagan and Sitharam 2009), Kolkata (Govindaraju and Bhattacharya 2012), Kanpur (Jishnu et al. 2013), Delhi (Mandal et al. 2014), the Mumbai Nuclear Power Plant and Port (Desai and Choudhury 2015), Mumbai city (Choudhury et al. 2015), Kolkata (Chatterjee and Choudhury 2016), Uttarakhand (Pandey et al. 2016), Dhanbad (Gupta et al. 2023), and Itanagar (Anshu et al. 2025a). Over time, the nonlinear method has advanced significantly, allowing for a more accurate representation of the nonlinear behavior of soil (Stewart and Kwok 2008). However, due to the complexity involved in selecting input parameters, nonlinear GRAs have been relatively rare worldwide. Some notable studies include New Madrid (Zheng and Luna 2011), Bam (Rayhani et al. 2008), Turkey (Kwok et al. 2008), Southern Switzerland (Roten et al. 2009), Chennai (Uma Maheshwari et al. 2010), Amingaon (Kumar et al. 2014), Guwahati (Dammala et al. 2017), and Bihar (Kumari et al. 2024). Several software packages are available for performing GRA. These tools estimate the tangent shear modulus, which characterizes the nonlinear behavior of soil, and solve the dynamic equation of motion in the time domain through step-by-step integration methods. Their primary distinction lies in how they simulate the nonlinear response and material damping of soils. Among the widely adopted nonlinear GRA programs are SUMDES-92 (Li et al. 1992), OpenSees (version 1.7.3) (Mazzoni et al. 2006), D-MOD 2000 (Matasović and Ordóñez 2007), DEEPSOIL (version 6.1), and TESS (Pyke 2000). Each of these programs is capable of performing 1D nonlinear GRA, while OpenSees and SUMDES can also simulate 2D and 3D ground shaking.

In recent times, DEEPSOIL has proven to be one of the most popular open-source software to conduct 1D equivalent-linear GRA. In order to perform such an analysis, the shear-wave velocity (or the equivalent small-strain shear modulus) and the unit weight of each soil layer are the important input parameters. The only other input once the profile of the soil and its properties are outlined is the input earthquake motion. Its acceleration record may be applied at a hypothetical bedrock outcrop, which is widely used as that is consistent with the assumptions of attenuation models or at the soil bedrock interface on the base of the column. The analysis output contains the time histories of shear stress, shear strain, and acceleration, along with their maximum values at the ground surface, the supposed outcrop of the bedrock and at various depths in the soil profile. A summary of previous studies using DEEPSOIL is presented in Table 4.

The amplification or deamplification of a region largely depends on the dynamic properties of the soil medium. Therefore, determining these properties is a crucial step in addressing geotechnical earthquake engineering challenges and conducting GRA studies. Moreover, dynamic soil properties are extensively used to evaluate the interaction between earthquake forces and the soil mass in the

selected area. Key dynamic properties include the shear modulus (G), modulus reduction ($G/G_{\max}$) curves, and the variation of damping ratio (ξ) with cyclic strains. Additional parameters essential for conducting GRA include the thickness of each soil layer, density, shear-wave velocity (V_s), and input ground motion. Among these, density plays a significant role in determining the low-strain shear modulus of subsurface soil layers. The study by Hanumantharao and Ramana (2008) classified the soils in their study into two broad groups: fine-grained and coarse-grained types. The essential density parameters were determined following the IS: 2720 Part 29 (BIS 1975) standard. The process of determining density in the field is time-consuming and may occasionally introduce minor errors in the measured values. Such slight variations in density can cause changes in the peak spectral value, typically ranging between 0.05 and 0.1g. However, these variations have a minimal effect on the estimated predominant period and spectral amplification (Hanumantharao and Ramana 2008). Consequently, assumed density values are rarely utilized for response analysis in exceptional situations.

An essential step in GRA is the selection of appropriate strain-dependent $G/G_{\max}$ and damping ratio (ξ) curves, which are critical for assessing the nonlinear response of the soil medium. When site-specific $G/G_{\max}$ and ξ curves are unavailable, researchers such as Anbazhagan and Sitharam (2008), Phanikanth et al. (2011), Reddy et al. (2021), Kumar and Krishna (2013), Shukla and Choudhury (2012a), and Kamatchi et al. (2008) have conducted GRA using globally developed $G/G_{\max}$ and damping ratio curves to estimate site amplification for various cities in India. Barani et al. (2013) examined the influence of $G/G_{\max}$ and ξ curves on site amplification and emphasized the importance of their selection in accurately determining the site amplification. Over time, researchers have developed numerous $G/G_{\max}$ and damping ratio curves tailored to different shear strain levels and material types. For various soil types, a wide range of $G/G_{\max}$ and damping ratio (ξ) curves is notably available in the literature, such as those proposed by Seed and Idriss (1970), Seed et al. (1986), Sun et al. (1988), Vucetic and Dobry (1991), the Electric Power Research Institute (EPRI 1993), Ishibashi and Zhang (1993), Rollins et al. (1998), Darendeli (2001), Menq (2003), Zhang et al. (2005, 2008), Kalligiorglou et al. (2008), and Chen et al. (2019). These curves are extensively used to describe the dynamic behavior of soil columns. The selection of inappropriate $G/G_{\max}$ and ξ curves can lead to significant errors, either underestimating or overestimating response parameters. Anbazhagan et al. (2017b) highlighted this issue by demonstrating how an improper curve selection could result in unreliable response analysis for deep soil sites, based on observations from KiK-Net data.

As soil is primarily a mixture of different size fractions; i.e., it being a combination of gravels, sands, silts, and clays, a proper soil classification is crucial to assess the dynamic properties. Understanding the behavior of such soil mixtures under dynamic loading conditions is essential for reliable analysis. Hanumanthrao and Ramana (2008) conducted cyclic triaxial tests on soil specimens prepared from mixtures of river Yamuna sand and nonplastic fines to evaluate the behavior of mixed soils. The study revealed that for soils without plastic fines, the $G/G_{\max}$ and damping ratio (ξ) curves aligned closely with the lower bound curves proposed by Seed and Idriss (1970). Anbazhagan et al. (2017b) suggested that the $G/G_{\max}$ and ξ curves provided by Seed and Idriss (1970) are appropriate for sandy soil sites, those by Rollins et al. (1998) are suitable for gravel sites, and EPRI (1993) curves are more applicable for rock sites. Similarly, Okur and Ansal (2007) investigated soil behavior following the 1999 Kocaeli earthquake and observed that the plasticity index plays a significant role in defining $G/G_{\max}$ and ξ curves, particularly for fine-grained soils, as highlighted by Vucetic and Dobry (1991).

Critical Appraisal and the Way Forward

This study goes beyond a conventional review by integrating PSHA and equivalent-linear GRA for Itanagar, one of the least studied yet most seismically vulnerable Indian cities. As amplification of motion in soft soil, at surface level, reaches 4.3 times higher than the bedrock level PGAs ranging from 0.18 to 0.36 g, it reinforces the necessity of seismic hazard classification of this region. These quantitative results, when set as benchmarks against the Indian urban cities such as Delhi, Gujarat, Kolkata, Manipur, Shillong, and Guwahati, reveal both the strength and the deficiencies of seismic hazard practices in Itanagar.

However, based on the existing studies, three critical insights have emerged: first, though the logic-tree model based global PSHA framework, Bayesian updating and region-specific GMPEs are in existence, and the hazard analysis for Itanagar is performed with an existing attenuation model of other regions and an incomplete earthquake catalogue due to the shortage of region-specific data, which might introduce epistemic uncertainties that weaken the reliability for local risk-sensitive planning. Second, though the nonlinear GRA provides the most precise design parameters due to realistic soil model, this study uses equivalent-linear GRA, which approximates the nonlinear soil behavior and provides a conservative result especially in soft soil and high seismic intensity. Thus, a nonlinear GRA must be carried out considering the actual strain-dependent dynamic soil properties, such as variations of $G/G_{\max}$ and damping ratio with shear strains. Moreover, Itanagar city consists of Quaternary deposits with steep slope, and the liquefaction potential may not be revealed due to a deep groundwater table; however, the experimental investigations need to be performed to understand the liquefaction potential of Quaternary deposits. Third, since the stiffness of actual ground conditions is of utmost importance in accurate hazard modeling and engineering application, a systematic site characterization using MASW, SPT, and SCPT needs to be carried out to understand the actual soil conditions below ground level and also to avoid the uncertainties that come with the consideration of V_{s30} .

The existing studies also revealed that the seismic hazard research has progressed for several metropolitan cities but still remains for several other seismically active and high-risk cities such as Itanagar. IS 1893:2016 has also recognized Itanagar as a very severe seismic hazard zone with large population density; hence the city may face severe seismic intensity in the near future, requiring a proper methodology and tools with a region-specific dataset for safe structural design from futuristic earthquakes.

Therefore, a future study must be conducted to assess the seismic hazard analysis considering a local or regional dataset, indicating seismic instrumentation, detailed geotechnical site characterization, the development of new GMPEs, and nonlinear ground response modeling, to make a safe and seismically resilient structural design framework, and thereby a proper foundation for resilient infrastructure planning and urban safety will be achieved. Moving forward, the adoption of Bayesian PSHA, logic-tree approaches for epistemic uncertainty, and machine learning driven ground-motion prediction models will enhance the robustness, adaptability, and global applicability of hazard assessment frameworks, thereby enhancing their role in data-driven strategies for seismic risk reduction. Additionally, seismic microzonation mapping should be seen as a continuous process rather than a completed task, since maps can always be refined and updated as newer and more accurate data become available.

Data Availability Statement

Data associated with this study are available from the corresponding author upon reasonable request.

Author Contributions

Aditya Kumar Anshu: Conceptualization; Methodology; Writing – original draft; Writing – review and editing. Jumrik Taipodia: Conceptualization; Funding acquisition; Project administration; Resources; Software; Supervision; Writing – review and editing. Shiv Shankar Kumar: Conceptualization; Writing – review and editing. Arindam Dey: Conceptualization; Supervision; Writing – review and editing.

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